

KESHAN, China

WMO: 506580

Lat: 48.0842N

Lon: 125.7880E

Elev: 292

StdP: 97.87

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-28.9	-34.4	0.2	-30.6	-32.2	0.2	-28.4	7.2	-15.6	6.3	-16.5	0.7	340	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	30.9	20.0	29.2	19.9	27.8	19.7	23.5	27.0	22.5	26.3	21.6	25.5	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.7	25.4	21.3	16.5	24.7	20.2	15.5	23.9	72.0	26.9	68.1	25.9	64.7	25.7	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.0	6.0	-33.1	34.0	3.7	2.0	-35.8	35.5	-37.9	36.7	-40.0	37.8	-42.7	39.3		
(5)			-33.2	24.9	3.7	1.1	-35.9	25.6	-38.0	26.3	-40.1	26.9	-42.8	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-20.8	-15.9	-5.8	5.5	13.8	20.2	22.5	20.2	14.0	4.5	-8.4	-18.6	(6)
(7)		DBStd	15.99	4.59	5.27	6.38	4.70	4.20	3.61	2.43	2.71	3.80	4.73	6.53	4.52	(7)
(8)		HDD10.0	3959	956	726	491	150	11	0	0	9	178	552	886		(8)
(9)		HDD18.3	6008	1214	959	749	386	153	22	2	137	428	802	1144		(9)
(10)		CDD10.0	1288	0	0	0	14	128	304	387	317	129	9	0	0	(10)
(11)		CDD18.3	296	0	0	0	1	11	76	131	71	7	0	0	0	(11)
(12)		CDH23.3	2240	0	0	0	6	152	687	903	426	66	1	0	0	(12)
(13)		CDH26.7	592	0	0	0	1	38	234	241	68	9	0	0	0	(13)
(14)	Wind	WSAvg	2.5	1.7	1.9	2.8	3.4	3.4	2.7	2.5	2.3	2.5	2.7	2.3	1.9	(14)
(15)	Precipitation	PrecAvg	522	3	3	7	21	38	90	165	106	57	18	7	6	(15)
(16)		PrecMax	858	11	16	29	79	90	274	353	264	122	41	30	19	(16)
(17)		PrecMin	333	0	0	0	2	6	24	14	37	4	0	0	0	(17)
(18)		PrecStd	135	3	3	7	19	22	51	85	57	36	12	8	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.0	2.1	12.5	23.4	30.2	34.7	33.2	30.1	27.9	20.7	9.4	-4.1	(19)
(20)			MCWB	-8.5	-1.8	4.9	11.0	15.2	19.5	21.7	21.6	18.4	12.5	4.5	-5.6	(20)
(21)		2%	DB	-9.9	-1.8	8.7	19.9	27.5	31.3	30.9	28.5	25.1	17.0	6.0	-7.8	(21)
(22)			MCWB	-11.0	-4.3	2.7	9.2	14.4	18.8	21.7	21.4	17.0	10.0	2.2	-8.9	(22)
(23)		5%	DB	-12.3	-5.0	5.9	16.9	24.7	29.3	29.2	27.2	23.2	14.4	2.9	-9.9	(23)
(24)			MCWB	-13.3	-6.9	1.1	7.8	13.2	18.4	21.6	20.8	15.8	8.7	-0.1	-10.9	(24)
(25)		10%	DB	-14.2	-7.7	3.4	14.1	22.1	27.3	27.7	25.9	21.1	12.1	0.4	-11.9	(25)
(26)			MCWB	-15.0	-9.2	-0.4	6.4	12.0	17.8	21.1	20.0	14.8	7.1	-1.9	-12.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.5	-1.2	5.8	12.1	17.1	22.3	25.3	24.4	20.3	13.7	4.7	-5.5	(27)
(28)			MCDWB	-7.2	1.5	11.2	20.4	26.1	28.6	28.6	27.1	25.8	17.8	8.1	-4.6	(28)
(29)		2%	WB	-11.0	-4.4	3.1	10.0	15.5	21.0	24.0	23.1	18.7	11.2	2.3	-8.9	(29)
(30)			MCDWB	-9.9	-1.8	7.9	18.1	23.7	27.6	27.6	26.5	22.8	15.6	5.8	-7.9	(30)
(31)		5%	WB	-13.2	-6.9	1.3	8.4	14.4	20.0	23.0	22.1	17.2	9.4	0.1	-10.8	(31)
(32)			MCDWB	-12.3	-5.1	5.3	15.9	22.4	26.2	26.6	25.7	21.1	13.2	2.5	-9.9	(32)
(33)		10%	WB	-15.0	-9.2	-0.2	6.9	13.2	19.0	22.1	21.1	15.8	7.5	-1.8	-12.7	(33)
(34)			MCDWB	-14.2	-7.8	3.1	13.1	20.6	24.8	26.0	24.5	20.0	11.8	0.3	-11.9	(34)
(35)	Mean Daily Temperature Range		MDBR	8.8	10.4	10.1	10.8	11.0	9.9	8.3	8.8	10.8	9.7	8.6	7.9	(35)
(36)		5% DB	MCDBR	9.6	12.0	12.3	15.2	15.0	13.1	10.9	10.6	13.8	13.2	11.5	9.3	(36)
(37)			MCWBR	8.7	10.0	7.8	8.2	6.5	4.9	4.4	4.8	6.4	7.7	8.2	8.4	(37)
(38)			MCDBR	9.6	11.7	11.3	13.7	12.5	10.3	8.1	8.4	10.7	11.5	11.0	9.1	(38)
(39)		5% WB	MCWBR	8.7	9.8	7.5	7.9	6.1	4.5	4.4	4.6	5.9	7.8	8.1	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.289	0.311	0.369	0.464	0.533	0.547	0.519	0.462	0.413	0.389	0.324	0.295	(40)	
(41)		taud	2.179	2.125	2.036	1.919	1.811	1.881	1.986	2.117	2.175	2.123	2.219	2.163	(41)	
(42)		Ebn at Noon	763	830	830	783	742	736	751	779	780	722	710	694	(42)	
(43)		Edh at Noon	82	109	141	176	204	192	171	143	123	109	79	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.92	3.10	4.40	5.06	5.66	5.83	5.44	4.98	4.17	2.88	1.80	1.43	(44)	
(45)		RadStd	0.12	0.19	0.31	0.41	0.46	0.67	0.57	0.38	0.33	0.17	0.14	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page